

Introduction to Euler–Maruyama Method

1 Introduction

The Euler–Maruyama (EM) method is a fundamental numerical technique used for simulating and approximating solutions to stochastic differential equations (SDEs). It is an extension of the classical Euler method for ordinary differential equations (ODEs) and accommodates the randomness inherent in SDEs.

SDEs are central to fields like quantitative finance, physics, biology, and engineering, where systems are influenced by both deterministic and stochastic factors. The EM method provides a practical tool for constructing sample paths of such systems.

2 Mathematical Formulation

2.1 General Form of an SDE

An Itô stochastic differential equation is generally written as:

$$dX_t = \mu(X_t, t) dt + \sigma(X_t, t) dW_t$$

where:

- X_t is the stochastic process of interest,
- $\mu(X_t, t)$ is the drift coefficient (deterministic component),
- $\sigma(X_t, t)$ is the diffusion coefficient (random component),
- W_t is a standard Wiener process (Brownian motion).

2.2 Discretized Time Grid

Let $t_0 < t_1 < \dots < t_N$ be a partition of the time interval $[0, T]$, with constant time step $\Delta t = t_{n+1} - t_n$.

3 Euler–Maruyama Discretization Scheme

Given the initial value $X_0 = x_0$, the EM scheme approximates the solution at each step by:

$$X_{n+1} = X_n + \mu(X_n, t_n) \Delta t + \sigma(X_n, t_n) \Delta W_n$$

where $\Delta W_n = W_{t_{n+1}} - W_{t_n} \sim \mathcal{N}(0, \Delta t)$, i.e., a Gaussian random variable with mean 0 and variance Δt .

4 Step-by-Step Algorithm

4.1 Inputs

- Final time T
- Number of steps N
- Initial value $X_0 = x_0$
- Drift function $\mu(x, t)$
- Diffusion function $\sigma(x, t)$

4.2 Procedure

1. Set $\Delta t = T/N$, define grid $t_n = n\Delta t$
2. Initialize $X_0 = x_0$
3. For $n = 0$ to $N - 1$:
 - Sample $\Delta W_n \sim \mathcal{N}(0, \Delta t)$
 - Compute:
$$X_{n+1} = X_n + \mu(X_n, t_n) \Delta t + \sigma(X_n, t_n) \Delta W_n$$

5 Example: Geometric Brownian Motion (GBM)

Consider the SDE:

$$dX_t = \mu X_t dt + \sigma X_t dW_t$$

with closed-form solution:

$$X_t = X_0 \exp \left(\left(\mu - \frac{\sigma^2}{2} \right) t + \sigma W_t \right)$$

Euler–Maruyama approximation:

$$X_{n+1} = X_n + \mu X_n \Delta t + \sigma X_n \Delta W_n$$

6 Applications

- Financial modeling (e.g., option pricing)
- Population dynamics in biology
- Stochastic control and filtering
- Particle systems in physics

7 Application to Derivative Pricing via Monte Carlo

In quantitative finance, the Euler–Maruyama method is often used to simulate asset price paths under risk-neutral dynamics. Once paths are generated, option pricing can be performed using Monte Carlo methods.

7.1 General Framework

1. Simulate M sample paths $\{S_T^{(i)}\}_{i=1}^M$ using Euler–Maruyama.
2. Compute option payoffs $f(S_T^{(i)})$ for each path based on the option type.
3. Take the average of payoffs: $\hat{P} = \frac{1}{M} \sum_{i=1}^M f(S_T^{(i)})$.
4. Discount to present value: $V_0 = e^{-rT} \cdot \hat{P}$.

7.2 Example: European Call Option

- Payoff: $f(S_T) = \max(S_T - K, 0)$
- Price estimate:

$$V_0 = e^{-rT} \cdot \frac{1}{M} \sum_{i=1}^M \max(S_T^{(i)} - K, 0)$$

8 References

- Kloeden, P. E., & Platen, E. (1992). *Numerical Solution of Stochastic Differential Equations*. Springer.
- Higham, D. J. (2001). An Algorithmic Introduction to Numerical Simulation of SDEs. *SIAM Review*.
- Glasserman, P. (2004). *Monte Carlo Methods in Financial Engineering*. Springer.